

RYAN R. HU

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EDUCATION

California Institute of Technology (Caltech)

B.S. Computer Science, Minor in Mathematics - GPA: 4.2/4.0

Pasadena, CA

Graduation Date: Jun 2027

Relevant Coursework: Computer Programming, Programming Methods, Software Design, Deep Learning, Learning Systems, Machine Learning & Data Mining, Algorithms, Computing Systems, Differential Equations, Mathematical Foundations of Computer Science (Discrete Mathematics), Probability and Statistics, Calculus of One and Several Variables and Linear Algebra

Teaching Assistant: Decidability and Tractability (Jan 2025 - Present)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript/TypeScript, HTML/CSS, SQL, MATLAB, $\LaTeX$

Developer Tools: Git, VS Code, AWS, Github, Linux, Docker, Kubernetes, Figma

Frameworks/Libraries: React.js, Node.js, Pandas, OpenCV, PyTorch, Keras, TensorFlow, Scikit, Tailwind, Flask, Plotly, PySpark

Skills: CI/CD, Machine Learning, Full Stack, Data Analysis, Debug Logging, Containerization, Unit Testing

EXPERIENCE

Machine Learning Researcher

Feb 2025 – Present

Caltech - Alvarez Lab

Pasadena, CA

- Building custom autoencoder architecture to detect anomalous player behavior in Activision's Call of Duty (COD) games, using SQL and Python to clean and process 3M+ raw gameplay logs, identifying 17,000+ anomalous players within a 10-day window through reconstruction error.
- Designing a verification pipeline integrating UMAP + HDBSCAN clustering with temporal segmentation, identifying distinct types of anomalous behaviors and analyzing their impact on player engagement in 200,000+ players.

Machine Learning Intern

Jun 2025 – Sep 2025

Tracevision - Math Team

Los Angeles, CA

- Automated data preprocessing pipelines and scripts to extract and construct 6000+ shot sequences from 80+ hours of raw basketball game footage centered around the basket, optimizing video format and quality for annotation teams through frame-level segmentation and noise reduction.
- Designed and deployed production-grade hybrid RNN + CNN shot detection model, achieving 88% accuracy on made vs. missed shot classification from annotated video clips.

Software Engineer Intern

Apr 2025 – Sep 2025

Tracevision - Math Team

Los Angeles, CA

- Improved ptools, a full-stack internal tool for object tracking and annotation review, by optimizing frontend and backend components to boost efficiency and enhance usability - cutting detection load latency from 1.2s to 0.7s (~ 40% faster). Built with Python, MySQL, HTML/CSS, and JavaScript.
- Developed geospatial API deployment pipelines using a combination of custom scripts and AWS infrastructure (EC2, S3, Lambda), streamlining integration workflows for 30+ facilities and 100+ cameras in security and retail sectors.

Undergraduate Researcher

Apr 2024 – Dec 2024

Caltech - Alvarez Lab

Pasadena, CA

- Analyzed HIV patient datasets collected by the Multicenter AIDS Cohort Study (MACS) by applying causal ML and gradient boosting trees on 800+ patients, discovering significant epigenetic markers correlated with accelerated aging - currently authoring manuscripts on findings.
- Built data preprocessing pipeline merging epigenetic datasets with web-scraped environmental satellite data, performing data cleaning, imputation, and feature transformation for machine learning tasks downstream.

Undergraduate Researcher

Jun 2023 – Sep 2023

Caltech - GALT Lab

Pasadena, CA

- Developed a visual anemometry framework in MATLAB combining YOLOv3 object detection (95.4% precision) with optical flow mapping techniques to estimate wind speeds within ± 5 m/s error.
- Conducted a quantitative analysis linking canopy motion variance to wind standard deviation across three different tree species, achieving >0.9 correlation in field tests.

SELECTED PROJECTS

Agentic Stock Trader | Python

- Built a multi-agent stock trading framework where LLM-driven agents ingest 1,000+ financial news items, debate with each other, and output explainable decisions, achieving up to 2.34% monthly return for various individual stocks in backtests.
- Designed a recursive agent debate architecture with static personas, dynamic leaf agents, and hierarchical clustering, creating 50+ distinct agents across 3 debate layers and multiple decision rounds to systematically evaluate trades and reach agreement.

Endless Runner Video Game | C, HTML/CSS, SDL

- Developed Google Dino Run-inspired endless runner game with obstacles, powerups, and progressive difficulty scaling. Written entirely in C using standard libraries.
- Implemented 100% unit test coverage for the physics engine. Reduced collision computation by ~40% overhead by restricting hitbox checks to active on-screen segments only.

ACTIVITIES/ACHIEVEMENTS

NCAA Division III Men's Basketball

Sep 2023 – April 2025

SCIAC All-Academic Team

2024, 2025